

Amritanshu Kumar

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Professional Summary

Builds game systems end to end — engine internals, real-time multiplayer, distributed backend — and leads the teams that ship them. Directs a 7-engineer team on a live title with **41K registered** / **~3.4K daily active** players, owning the server and API architecture. Built a C++ physics engine and a Unity hot-reload tool (**7ms** IL patching) from scratch. **5M+ downloads** shipped.

Professional Experience

RootHoot Pvt Ltd

Remote, Hyderabad

Team Lead

Dec 2024 – Present

- Own the server-side architecture end to end for Pongz, a live multiplayer Mahjong title shipping on Android and iOS — TypeScript game server, Express REST API, and cloud infrastructure.
- Scaled the game server horizontally from a single instance** — Redis pub/sub adapter for cross-instance event fan-out, leader election to form matches from a shared global queue (fixing matchmaking split-brain), distributed locking, and failover that hands live games to peer nodes on drain with zero session loss.
- Architected a unified connection layer to replace fragile ad-hoc socket handling — state machine for lifecycle management, typed event dispatch, offline queue with TTL expiry, and adaptive connectivity probing.
- Engineered a server-authoritative protocol that eliminated race conditions from latency and message reordering — monotonic turn sequencing, fingerprint-based dedup on reconnect, and exponential backoff with jitter.
- Designed a three-stage AFK system to keep matches running when players drop — warning, server-tracked idle, then bot substitution. Full state resync on return without disrupting the other three.
- Replaced scattered boolean flags with a hierarchical state controller (8 phases, 14+ substates) that became the single source of truth for gameplay, UI, and networking.
- Extracted 8 core systems into shared team infrastructure — Event Bus, State Machine, Save, Pause, Logging, UI, Loading, Popup — with separated event/message buses, WeakReference lifecycle management, atomic-write persistence, and bit-flag pause layers.
- Grew the test suite to **3,332 passing tests** with CI-enforced coverage floors (80% lines / 85% functions), including real-socket and multi-instance E2E harnesses.
- Directed architecture, code reviews, and sprint planning across client, server, and API while staying hands-on.

DeftSoft

Mohali, Punjab

Game Developer

Feb 2023 – Oct 2024

- Shipped an endless runner to **5M+ downloads** and **4.6 stars** on Google Play. Iterated on player data post-launch to push engagement up 30%.
- Introduced TDD across the team to catch regressions before release — ratings improved **1.5 stars** as a direct result. Authored a **25-chapter** technical course on game architecture.
- Built procedural tools to speed up content creation — cut level-design time by 40%. Optimized rendering to hit 25% higher FPS on low-end devices.
- Mentored 3 junior developers — code reviews, pair programming, and coding standards.

Selected Projects

InstaReload: Unity hot-reload tool that cuts domain reload from 3–5s to **7ms**, removing the main bottleneck in the edit-test loop. Two paths: Roslyn for debug edits (90% of cases), MonoMod detours for structural changes.

Nimirta Engine: Custom 2D engine, C++/SFML — impulse solver with warm starting, Baumgarte correction for stable high-velocity contacts, semi-implicit Euler, fixed timestep with spiral-of-death prevention, 3 collision algorithms, 8 materials, hand-written vector math, 3-tier AI.

FlowUI: Open-source Unity UI framework (MIT) that eliminates UI boilerplate — CRTP builders, Roslyn codegen with partial classes, bidirectional O(1) lookups, and an 8-tab visual editor. Fully documented.

Technical Skills

- Languages:** C#, TypeScript, C++, Go, Python
- Game Engines:** Unity, Unreal Engine, Custom (C++)
- Backend:** Node.js, Express, MongoDB, Redis, Socket.IO, Docker, Google Cloud (App Engine), Firebase
- Specializations:** System Architecture, Real-Time Multiplayer, Horizontal Scaling, Leader Election & Distributed Locking, Physics Simulation, State Synchronization, TDD, ECS

Education & Certifications

Bachelor of Computer Application, Lovely Professional University, Punjab, India | Sep 2020 – Jun 2023

Certifications: Unity Certified Programmer | Unreal Engine Game Design & Development Specialization